

Lecture 5 Review Handout

Statistics 367-1-4361

Lecture Overview

Lecture 5 moves from simple coin-flip models to data with **group structure**, using hospital death rates as the running example. The lecture compares pooled, unpooled, and hierarchical models, showing how hierarchical models share information across groups while still allowing group-level differences. It then uses the Bayesian workflow to fit and check a hierarchical beta-binomial model, interpret shrinkage, diagnose funnel geometry, and extend the same logic to multi-level models. The lecture ends by summarizing both the benefits and the pitfalls of hierarchical modeling.

Lecture Sections

5.1 — Grouped data

This section introduces the problem that motivates the whole lecture: most biomedical data are not just a single flat sample. Measurements are often grouped by hospital, subject, session, animal, cell culture, treatment, prosthesis, day, or trial. The hospital cardiac-arrest example shows the central question: apparent differences between hospitals may reflect real structure, sampling noise, or both.

New Definitions and Terms

- **Grouped data** — data in which observations are organized into meaningful groups, such as hospitals, subjects, sessions, or treatment groups
- **Group structure** — the organization of data into levels that may affect how observations should be modeled

Figure

Do hospitals have different death rates?

- Differences seem large
- But maybe they are just noise

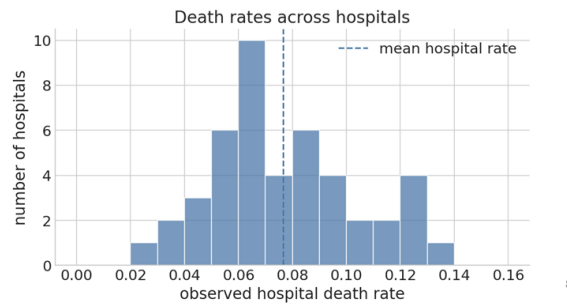


Figure 1: Slide 8 — Do hospitals have different death rates?

5.2 — Pooled model

A pooled model collapses the hospital structure and estimates one death-rate parameter from all cases together. This gives a simple baseline model: one probability of death is used for all hospitals. The workflow then asks whether the prior predictive data are plausible and whether the posterior predictive data reproduce the variability seen across hospitals.

New Definitions and Terms

- **Pooled model** — a model that ignores group structure and estimates a single parameter from all observations together
- **Pooling** — combining data across groups so that the groups do not have separate parameters

Formulas

$$\theta \sim \text{Beta}(\alpha_{\omega}, \beta_{\omega})$$

The pooled model has one death-rate parameter, shared across all hospitals.

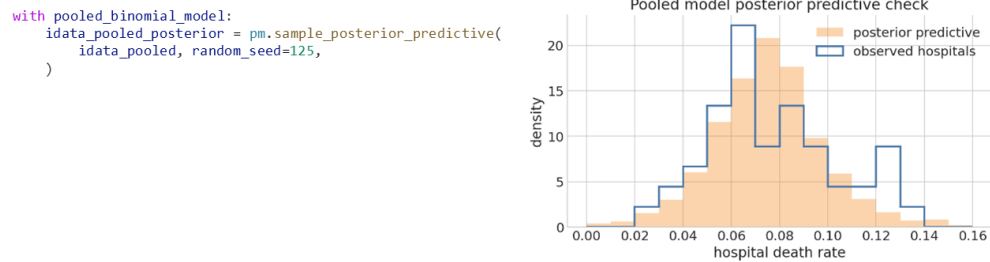
$$z_i \sim \text{Binomial}(n_i, \theta)$$

Each hospital's observed deaths are modeled using the same probability parameter.

Figure

Check posterior predictive for model fit

- Real hospitals seem more variable than posterior predictive hospitals



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Figure 2: Slide 14 — Check posterior predictive for model fit

5.3 — Unpooled model

An unpooled model gives each hospital its own probability parameter. This captures hospital-to-hospital differences, but it does not allow hospitals to inform each other. The posterior plots show the resulting increase in variability and uncertainty for individual hospital parameters.

New Definitions and Terms

- **Unpooled model** — a model that estimates separate parameters for each group without sharing information across groups
- **No information sharing** — a modeling structure in which data from one group do not affect the parameter estimates for other groups

Formulas

$$\theta_i \sim \text{Beta}(\alpha_\omega, \beta_\omega)$$

The unpooled model assigns a separate death-rate parameter to each hospital.

$$z_i \sim \text{Binomial}(n_i, \theta_i)$$

Each hospital's observed deaths depend only on its own parameter.

Figure

Pooled and unpooled posterior plots

- Posterior plots show the increased variability
 - No information is shared across the hospitals

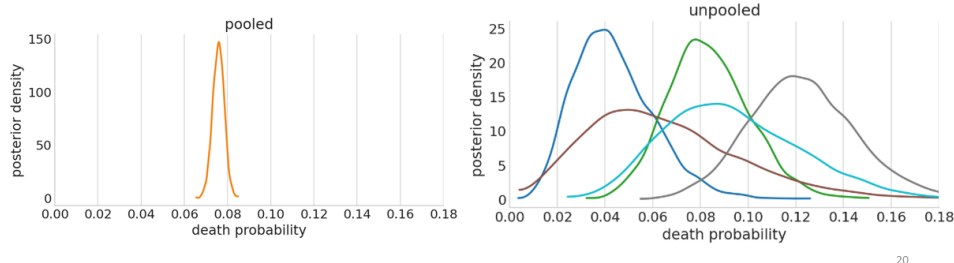


Figure 3: Slide 20 — Pooled and unpooled posterior plots

5.4 — Hierarchical model

The hierarchical model makes the parameters of the hospital-level prior into unknown quantities. These new parameters are estimated from the data, so each hospital parameter is influenced by its own data and by the other hospitals. This produces information sharing while still preserving hospital-level differences.

New Definitions and Terms

- **Hierarchical model** — a model in which parameters for lower-level groups are themselves modeled as coming from a higher-level distribution
- **Hyperparameter** — a parameter of a prior distribution, such as the population-level mean or concentration governing hospital-level parameters
- **Hyperprior** — a prior distribution placed on a hyperparameter
- **Information sharing** — the way group-level parameter estimates are influenced by data from other groups through shared hyperparameters

Formulas

$$\mu_\theta \sim \text{Beta}(\mu_{\mu_\theta}, \nu_{\mu_\theta})$$

The overall mean death rate is treated as an unknown hyperparameter.

$$\nu_\theta \sim \text{HalfNormal}(\sigma_{\nu_\theta})$$

The similarity of hospital death rates is also treated as an unknown hyperparameter.

$$\theta_i \sim \text{Beta}(\mu_\theta, \nu_\theta)$$

Hospital-specific death rates are drawn from a shared population distribution.

Figure

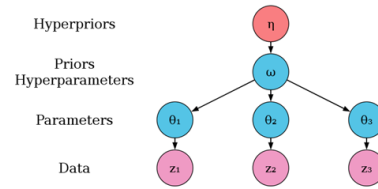
The hierarchical model causes information sharing

$$\omega \sim p(\eta)$$

$$\theta_g \sim p(\omega) \text{ for } g \in [1 \dots 4]$$

$$y_{g,i} \sim p(\theta_g)$$

- Posterior of θ_g depends on y and ω
- Posterior of ω depends on θ_g and $p(\eta)$
- Practically speaking:
 - Each θ_g is now influenced by all θ_g
 - Each θ_g is informed by more data



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Figure 4: Slide 25 — The hierarchical model causes information sharing

5.4 — Parameterizing Beta

This section shows that the same Beta distribution can be written using different parameterizations. The lecture first connects the Beta distribution to successes and failures, then rewrites it in terms of a mean and a concentration, and finally in terms of a mean and width. The hierarchical model uses the mean/concentration parameterization because it separates the center of the hospital distribution from the similarity of hospitals.

New Definitions and Terms

- **Concentration** — a Beta-distribution parameter controlling how tightly values cluster around the mean
- **Mean/concentration parameterization** — a way of writing the Beta distribution using a mean parameter and a concentration parameter
- **Mean/width parameterization** — a way of writing the Beta distribution using a mean parameter and a spread parameter

Formulas

$$\text{Beta}(\alpha, \beta)$$

The familiar parameterization treats α as success-like and β as failure-like.

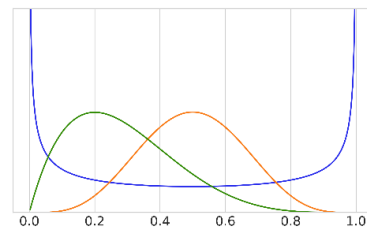
$$\text{Beta}(\mu, \nu)$$

The lecture uses μ as the mean and ν as the concentration.

Figure

Parameterizations of the Beta distribution

- As a prior equivalent to a number of successes and failures
 - α successes, β failures
- As a prior equivalent to number of attempts and mean success rate
 - μ mean, ν number of attempts
- By the mean and the width
 - μ mean, σ standard deviation
- We will use μ, ν



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Figure 5: Slide 30 — Parameterizations of the Beta distribution

5.5 — Coding hierarchical models

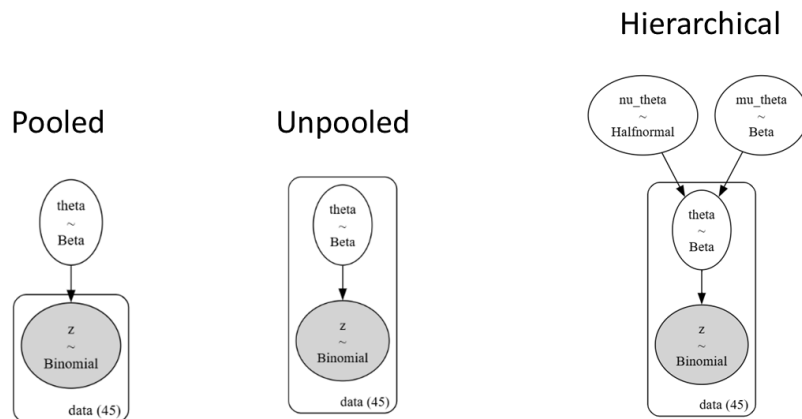
This section translates the graphical hierarchy into PyMC code. The data likelihood sits at the bottom, hospital-level parameters sit above the data, and hyperparameters with hyperpriors sit above the hospital-level parameters. The code also shows how the same model can be converted into a graphical model, making the hierarchy visible.

New Definitions and Terms

- **Model hierarchy** — the layered structure from data, to group-level parameters, to hyperparameters, to hyperpriors
- **Graphical hierarchy** — a model diagram that makes the levels of a hierarchical model explicit

Figure

Hierarchical models have an extra layer



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Figure 6: Slide 35 — Hierarchical models have an extra layer

Examine prior predictive

- Initial guess was too wide

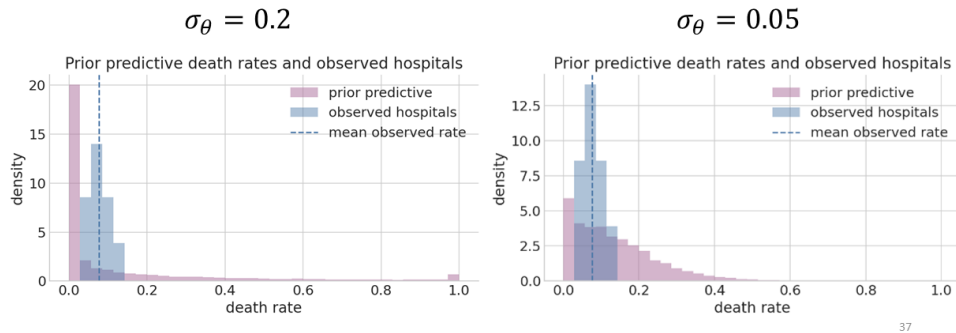


Figure 7: Slide 37 — Examine prior predictive

5.6 — Fitting hierarchical models

This section applies the workflow after fitting the hierarchical model. The lecture checks summary diagnostics, trace plots for hyperparameters, trace plots for selected hospital parameters, pair plots, divergences, and posterior predictive simulations. Only after these checks look reasonable is the posterior treated as interpretable.

New Definitions and Terms

- **Pair plot** — a plot of joint posterior draws for two parameters, used both for interpretation and for diagnosing difficult posterior geometry
- **Joint draw** — one posterior draw containing a self-consistent set of values for all parameters in the model

Figure

Pair plots show that parameters are drawn jointly

- Every dot is one draw
 - Each draw contains all parameters
 - Parameters are self-consistent
 - Within a draw
- Pair plots are also used for diagnosis
 - Divergences can reveal problematic regions of the posterior
 - Strong correlations may mean the model is not well specified

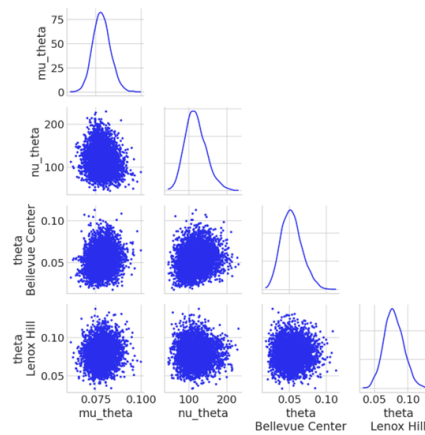
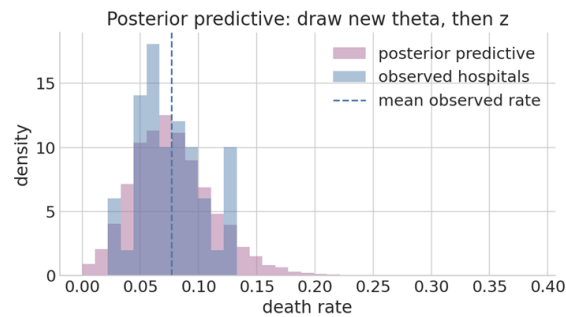


Figure 8: Slide 43 — Pair plots show that parameters are drawn jointly

Posterior predictive data is similar to real data

- For every sample:
 - Use sampled μ_θ, ν_θ
 - Draw new θ_i
 - For each new θ_i
 - Draw a new z_i



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Figure 9: Slide 44 — Posterior predictive data is similar to real data

5.7 — The posterior distribution

This section interprets the fitted hierarchical posterior. Compared with the pooled model, the hierarchical model has less certainty about the population mean because the mean reflects variation across hospitals, not just thousands of individual cases. Compared with the unpooled model, hospital-specific estimates are more certain and more similar because the hospitals inform one another through the hyperparameters.

New Definitions and Terms

- **Shrinkage** — movement of group-level estimates toward the population mean caused by information sharing in a hierarchical model
- **Regression toward the mean** — the tendency for extreme group estimates to move closer to the overall mean when group-level uncertainty is modeled
- **HDI width shrinkage** — reduction in posterior interval width, especially for groups with less data, due to hierarchical information sharing

Figure

Shrinkage is regression towards the mean

- The estimate of death rate in each hospital is closer to the mean across hospitals than would be expected from the data alone

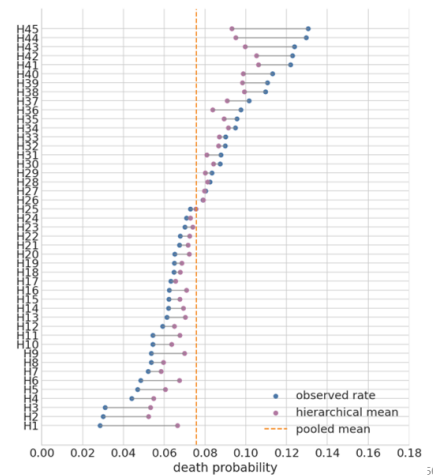
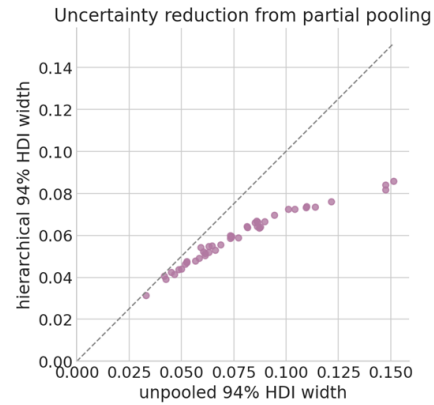


Figure 10: Slide 50 — Shrinkage is regression towards the mean

Hierarchical models also reduce uncertainty

- HDI width of the parameters shrinks
- The effect is stronger where there is less data



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Figure 11: Slide 51 — Hierarchical models also reduce uncertainty

5.8 — Funnels

Hierarchical models can produce difficult posterior geometry. A funnel appears when an unknown scale parameter controls how widely unknown location parameters can vary. When the scale becomes very small or very large, the shape of the posterior can become hard for NUTS to explore with one step size.

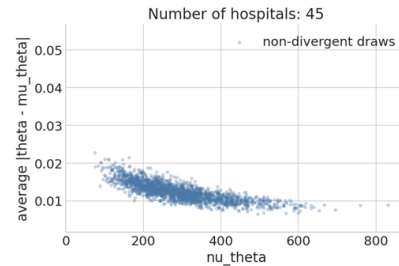
New Definitions and Terms

- **Funnel** — posterior geometry in which changes in a scale parameter strongly change the available range of other parameters
- **Location parameter** — a parameter describing the center of a distribution, such as a mean, median, or mode
- **Scale parameter** — a parameter describing the spread of a distribution, such as a standard deviation, variance, or interquartile range
- **Centered parameterization** — a hierarchical parameterization in which group parameters are drawn directly from a distribution controlled by unknown hyperparameters
- **Non-centered parameterization** — a reparameterization that separates standardized group variation from the unknown scale parameter

Figure

Our hierarchical model has a funnel

- As ν_θ gets large
 - θ get closer and closer together



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Figure 12: Slide 55 — Our hierarchical model has a funnel

When to worry about funnels

- An unknown **scale** is used as the scale of an unknown **location**
- The **scale** parameter can go to 0
 - The data is weak
 - The prior allows 0

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Figure 13: Slide 59 — When to worry about funnels

5.9 — Multi-level models

This section extends the same hierarchical logic to models with more than one level of grouping. The hospital data can be grouped by borough or by hospital type, and the slides build a borough-level model step by step: likelihood, hospital parameters, borough hyperparameters, and hyperpriors. The key idea is that Bayesian models can represent nested structure rather than forcing all variation into a single level.

New Definitions and Terms

- **Multi-level model** — a hierarchical model with more than one level of group structure
- **Nested hierarchy** — a structure in which lower-level units belong to intermediate groups, which themselves belong to a higher-level population
- **Group index** — an index variable used in code to connect each observation or group-level parameter to the correct higher-level group
- **Grand mean** — the population-level mean governing the distribution of group-level means

Formulas

$$\mu_{\theta,g} \sim \text{Beta}(\mu_{\mu_{\theta}}, \nu_{\mu_{\theta}})$$

Each borough has its own mean death rate, drawn from a higher-level distribution.

$$\nu_{\theta,g} \sim \text{HalfNormal}(\sigma_{\nu_{\theta}})$$

Each borough can also have its own within-borough similarity parameter.

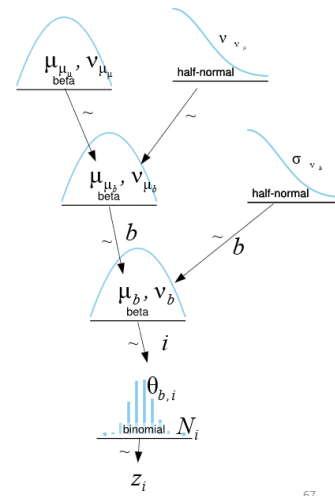
$$\theta_i \sim \text{Beta}(\mu_{\theta,g[i]}, \nu_{\theta,g[i]})$$

Each hospital's death rate is tied to the borough to which the hospital belongs.

Figure

Model step 4b: hyperprior 2

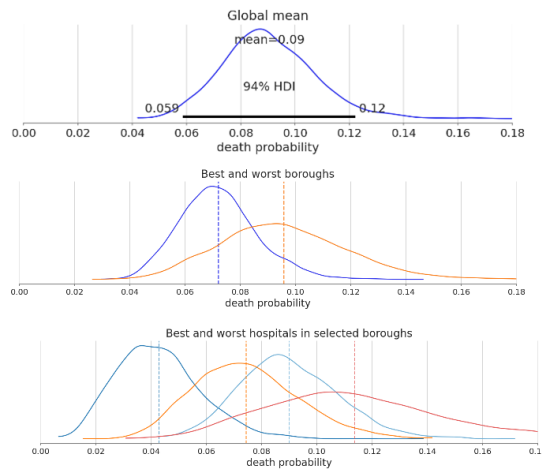
- Each hospital
 - Deaths and rate
- Each borough
 - Distribution of rates
- Across boroughs:
 - Distribution of mean rates
 - Mean across boroughs: μ_{μ_b}
 - Similarity of borough means: v_{μ_b}
 - Hyperpriors for both of these
 - Distribution of similarity within borough
 - Hyperprior on this



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Figure 14: Slide 67 — Model step 4b: hyperprior 2

Comparing population, position and individual means



Death rate is around 9%

The Bronx is around 7%
Staten Island around 9.5%
But there is less data on
Staten Island

Each borough has a range of
hospitals

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Figure 15: Slide 71 — Comparing population, position and individual means

5.10 — Summing up / looking forward

The lecture closes by emphasizing the tradeoff in hierarchical modeling. Hierarchical models can represent structured data, balance population and group-level information, and provide more robust estimates when some groups have little data. The cost is greater computational and interpretational complexity, including a stronger need for prior predictive checks, sampling diagnostics, posterior predictive checks, and careful multi-level interpretation.

Figure

Benefits and pitfalls of hierarchical models

Benefits

- Model structured data
- Balance
 - Population information
 - Group information
- Robust estimation
 - Even if some groups have less data

Pitfalls

- Computational complexity
 - Increased fitting time
 - Importance of sampling diagnostics
- Priors disconnected from data
 - Importance of prior predictive
- Interpretational complexity
 - Simultaneous interpretation at multiple levels

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Figure 16: Slide 74 — Benefits and pitfalls of hierarchical models